www.csvtuonline.com

Et 8 sem 3

Printed Pages – 7

328872 (28)

BE (8th Semester)

Examination, April-May, 2013

Branch : Et&T

RADAR ENGINEERING & NAVGATIONAL AIDS (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) is compulsory.

(ii) Attempt any two parts from (b), (c) & (d).

(iii) Answer should be strictly to the point.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

UNIT - I

Q. 1. (a) Define range resolution. 2

(b) What do you understand by range of radar ? Determine the maximum unambigious range and range resolution of a pulse radar having pulse width of 5 μs at a PRF of 1000 hz. 7

(c) Derive an expression to find the probability of false alarm in case of radar. Also discuss the false alarm time. 7

(d) A radar operating at 11 GHz with the peak power of 500 kW. The power gain of antenna

www.csvtuonline.com

www.csvtuonline.com

is 5000 and minimum power of receiver is 10^{-14} W. Calculate the maximum range of radar if the effective area of antenna is $10m^2$ and radar cross section is $4m^2$. 7

UNIT - II

Q. 2. (a) Explain the function of COHO in case of MTI radar. 2

(b) Explain the following terms : 7

(i) Blind Speeds

(ii) Staggered PRF

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) What do you understand by tracking of radar ? Explain in detail about monopulse tracking with the help of a block diagram. 7

(d) Explain delay line canceller. Derive an expression for output response of a MTI receiver with delay line canceller. 7

UNIT - III

Q. 3. (a) Explain cosmic noise. 2

(b) What do you understand by the term 'Ducting' ? Explain elevated duct and evaporation duct. 7

www.csvtuonline.com

www.csvtuonline.com

(c) Discuss the following in detail : 7

(i) Atmospheric absorption noise

(ii) Anthropogenic noise

(d) Explain the phenomenon of diffraction & refraction in case of propagation of radar waves. 7

UNIT - IV

Q. 4. (a) Mention two important functions of a radar antenna. 2

(b) Explain directive gain & power gain. Also discuss how they are related with each other ? 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Explain the working of cassegrain antenna with the help of suitable diagram. 7

(d) Write short notes on : 7

(i) Parabolid

(ii) Radomes

UNIT - V

Q. 5. (a) What do you understand by receiver noise figure ? 2

(b) Explain the working of balanced duplexer with the help of a neat sketch in both transmit and receive mode. 7

www.csvtuonline.com

www.csvtuonline.com

(c) Explain the working principle of Klystron.

Also comment about its bandwidth. 7

(d) Discuss in detail various radar displays. 7

www.csvtuonline.com

www.csvtuonline.com